

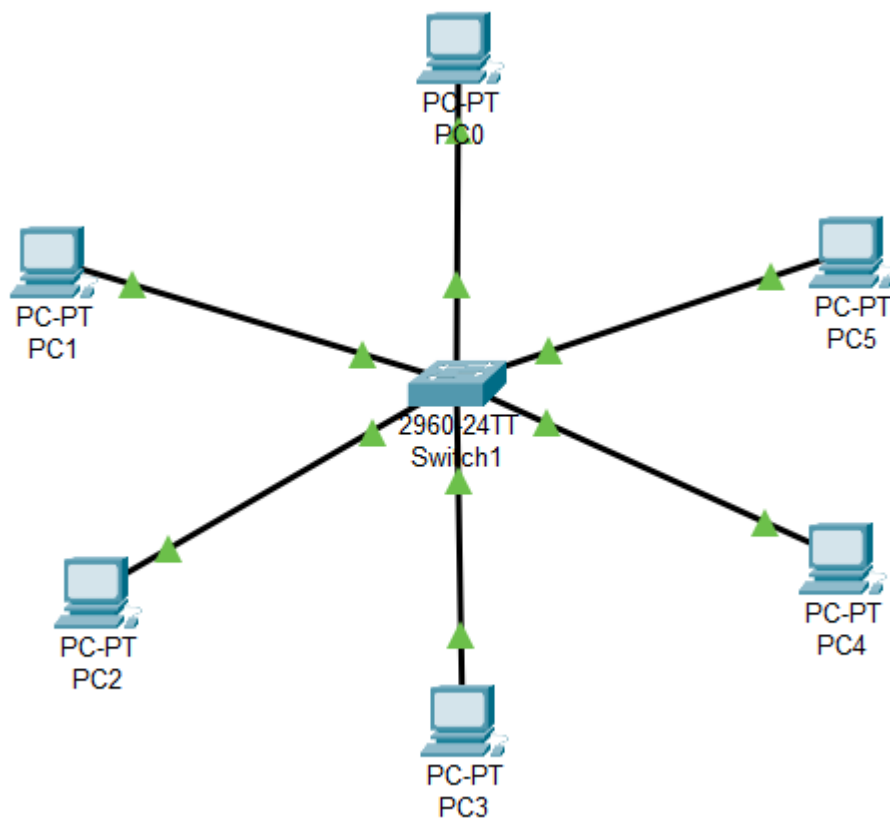
## Computer Networks: Assignment-3

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**Admission Number:** I24AI012

Q 1) Construct a star topology in the Cisco packet tracer as shown below, and check where the data packets are transferring from one system to another using ping.

**Diagram:**



**IP Config:**

**PC0:** 192.168.1.1 / 255.255.255.0

**PC1:** 192.168.1.2 / 255.255.255.0

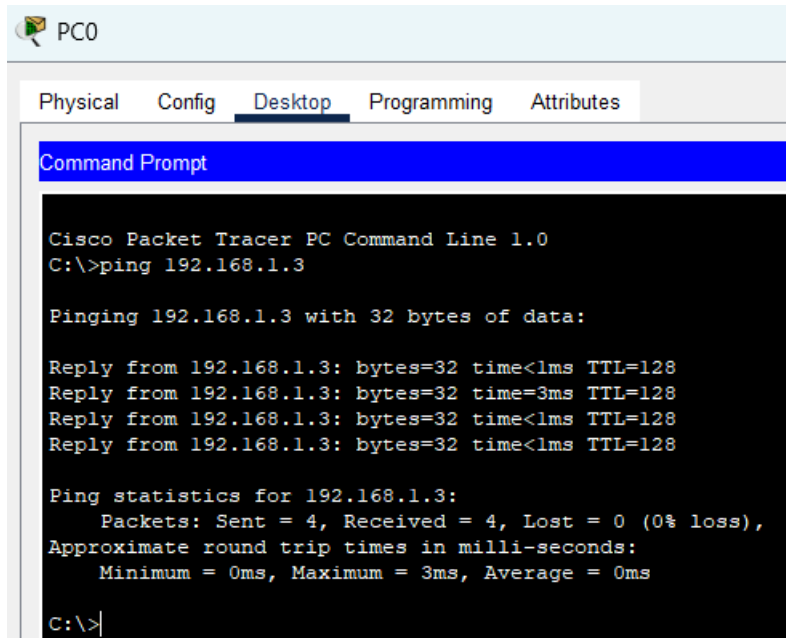
**PC2:** 192.168.1.3 / 255.255.255.0

**PC3:** 192.168.1.4 / 255.255.255.0

**PC4:** 192.168.1.5 / 255.255.255.0

**PC5:** 192.168.1.6 / 255.255.255.0

**Output:**



The screenshot shows the 'PC0' window in Cisco Packet Tracer. The 'Desktop' tab is selected, displaying a 'Command Prompt' window. The text in the Command Prompt is as follows:

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.3

Pinging 192.168.1.3 with 32 bytes of data:

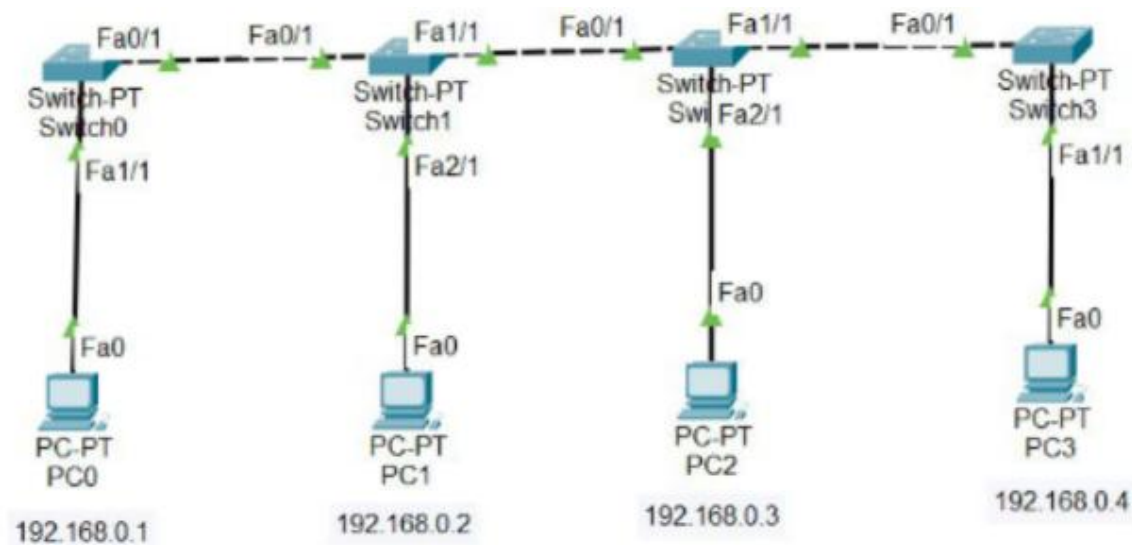
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time=3ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128
Reply from 192.168.1.3: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.3:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 3ms, Average = 0ms

C:\>|
```

Q2) Construct a bus topology in the Cisco packet tracer as shown below, and check where the data packets are transferring from one system to another.

**Diagram:**



**IP Config:**

**PC0:** 192.168.0.1 / 255.255.255.0

**PC1:** 192.168.0.2 / 255.255.255.0

**PC2:** 192.168.0.3 / 255.255.255.0

**PC3:** 192.168.0.4 / 255.255.255.0

**Output:**

```
PC0
Physical Config Desktop Programming Attributes
Command Prompt
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.0.4

Pinging 192.168.0.4 with 32 bytes of data:

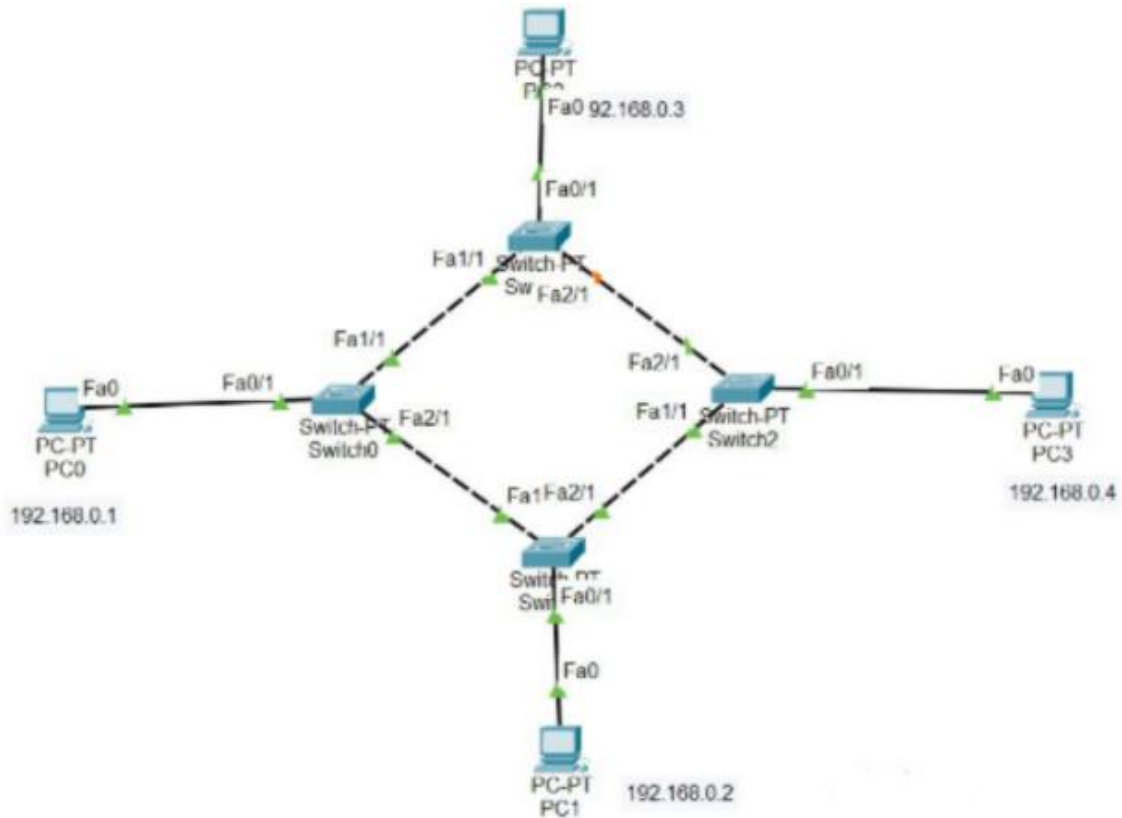
Reply from 192.168.0.4: bytes=32 time<1ms TTL=128
Reply from 192.168.0.4: bytes=32 time<1ms TTL=128
Reply from 192.168.0.4: bytes=32 time<1ms TTL=128
Reply from 192.168.0.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Q 3) Construct a ring topology in the Cisco packet tracer as shown below, and check where the data packets are transferring from one system to another.

**Diagram:**



**IP Config:**

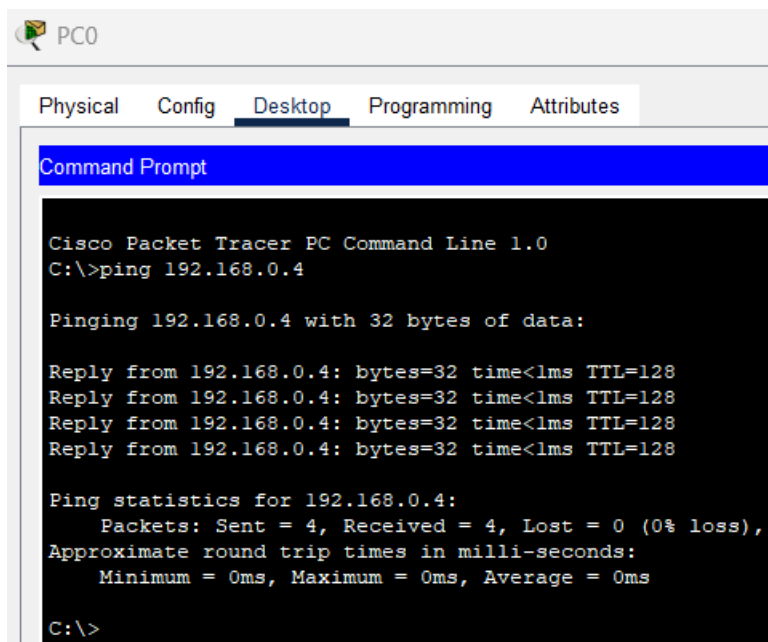
**PC0:** 192.168.0.1 / 255.255.255.0

**PC1:** 192.168.0.2 / 255.255.255.0

**PC2:** 192.168.0.3 / 255.255.255.0

**PC3:** 192.168.0.4 / 255.255.255.0

## Output:



The screenshot shows a Cisco Packet Tracer PC Command Line interface for PC0. The 'Desktop' tab is selected. The command prompt displays the output of a ping command to 192.168.0.4, showing four successful replies with 0% loss and 0ms round trip times.

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.0.4

Pinging 192.168.0.4 with 32 bytes of data:

Reply from 192.168.0.4: bytes=32 time<1ms TTL=128
Reply from 192.168.0.4: bytes=32 time<1ms TTL=128
Reply from 192.168.0.4: bytes=32 time<1ms TTL=128
Reply from 192.168.0.4: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.0.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```